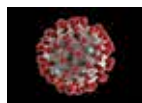




Tracking the spread

An important application of AI during the outbreak is to track how the virus spreads across populations. <https://digit.in/jul20-71>



Mundane applications

AI can be used for simple applications such as processing healthcare claims or as chatbots. <https://digit.in/jul20-76>

built a model which estimates the number of COVID-19 infections that have gone undetected by using machine learning and teaming up with the AWS Diagnostic Development Initiative. An MIT team has trained a ML (machine learning) model to accurately predict the spread of coronavirus using publicly-available data, while also taking into consideration factors such as social distancing, quarantine, and more being enforced by various governments.

AI can also help estimate how long a place should be kept under lockdown, and how much will be needed in the way of supplies, etc.

DIAGNOSIS, RESEARCH AND TREATMENT

Beyond simply predicting the spread of the virus, AI is also being used to diagnose and detect coronavirus in patients. A Chinese tech company, Alibaba claims that it can detect the coronavirus in a patient in just 20 seconds with an accuracy of 96%. It does so using the help of AI in CT scans. The algorithm, which was built in Damo Academy, was made using data from more than 5,000 confirmed cases, and use that data is able to detect patients through CT scans.

Along the same lines, AI-based startup Qure.ai is also assisting hospitals with their AI software which analyses X-rays and CTs, similar to what Alibaba was doing. The software itself has been in the works for 3 years, but with the pandemic breaking out, it's being adopted at a much faster rate. One of its advantages is that it can be deployed remotely, and is not impeded by the lockdown. It can also be taken door-to-door to conduct tests on-site.

In order to further research surrounding COVID-19, AWS has launched CORD-19 Search, which is a website powered by machine learning. Using the site, researchers can quickly search for papers and documents related to the pandemic, without needing to rifle through the increasing volume of information, inaccurate or otherwise, that's piling up online. The dataset for CORD-19 includes over

1,28,000 research papers and other material from around the world.

AI is also being used to find new ways to detect and diagnose the disease in patients. Recently, in April, three biotech students and a professor from Mumbai, claimed that an AI-enabled voice tool they're working on can diagnose COVID-19 through voice, using a smartphone. The tool was actually picked up for a pilot run by a Univer-

In a similar vein, Google's DeepMind AI is also helping with drug development, by studying the proteins that make up the virus.

CREATION OF PROTECTIVE GEAR

Sonovia, an Israeli startup, is using their AI-based antipathogen, antibacterial fabric which uses metal-oxide nanoparticles to create protective face masks for frontline workers.



Qure.ai uses X-rays and CTs to detect the presence of coronavirus

sity in Rome, where one of the biotech students, Priya Garg, claims that the tool is fully functional and in-use in Italy to detect patients.

The tool breaks down voice into multiple parameters, such as distortion and noise frequency. There are 6300 such parameters and human ears usually can't distinguish between the majority of them. However, the AI can. It compares these parameters to the values of a healthy person, and applies a patented technique to determine if the patient is positive or negative.

UK-based AI company BenevolentAI is using its platform to help accelerate the search for a cure or drug to treat coronavirus. After studying how the virus affects our bodies and how we respond to it, they used their AI to identify approved drugs which could help curb symptoms, and stop progression of the virus. Furthermore, the AI was also used to learn the circumstantial nature between the drugs, genes and the virus, to further reduce the sample size of possible drug compounds. The drug discovered by the AI, Baricitinib, a treatment for arthritis, is currently in clinical trials to discover its effectiveness as a potential treatment.

Hangzhou-based startup, Rokid Corp, have created AI-enabled glasses which can detect the temperature of visitors with no contact. The moment the glasses detect high temperature, it immediately sends alerts and makes a digital record.

OUTSIDE OF IMMEDIATE HEALTHCARE

Outside of directly helping detect and stop the virus, and finding treatment, AI is being used to help treatment actually reach those who need it via unmanned autonomous drones. Furthermore, they're helping ease the burden on the healthcare industry currently by helping with management of finances and accounts. Of course, they're also helping reduce the number of needed face-to-face interactions between patients and hospitals. Furthermore, AI chatbots are helping spread awareness about the disease while also curbing the spread of false information, which is just as dangerous as the virus itself in these times. A lot of the changes that have come about in the world of healthcare are very likely here to stay, even once this pandemic blows over, whenever it does. 